

Predicting Used Car Prices with Machine Learning: A Comparative Study of Models and Preprocessing Strategies

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Abstract

Used car price prediction is a practically important regression task that requires careful handling of heterogeneous tabular data. In this paper, we investigate the effectiveness of machine learning methods for predicting second-hand vehicle prices using the UK Used Car Dataset, comprising approximately 100,000 listings across 11 manufacturers. We address four research questions concerning prediction accuracy, model selection, the impact of preprocessing strategies, and feature importance. We compare five models spanning high-bias to low-bias: Ridge Regression, K-Nearest Neighbours, Decision Tree, Random Forest, and a Multi-Layer Perceptron. Our experiments demonstrate that the ensemble tree-based method substantially outperforms the linear baseline, with Random Forest reducing held-out test RMSE by 25.6% relative to Ridge Regression. The strongest preprocessing effects come from log-transforming the skewed price target and scaling distance-sensitive or regularised models, while transmission type, vehicle year, engine size, age, and mileage-related features provide the dominant predictive signals.

1 Introduction

The second-hand car market is one of the largest consumer goods markets globally, with millions of transactions per year. Accurately estimating a fair market price benefits both buyers—who risk overpaying—and sellers—who risk underpricing their vehicles. Traditional valuation relies on domain expertise and lookup tables, but these approaches fail to capture the complex, non-linear interactions among vehicle attributes such as brand, age, mileage, fuel type, and transmission. Machine learning offers a data-driven alternative that

can model these interactions directly from historical listing data (Gegic et al., 2019).

This project investigates machine learning methods for used car price prediction using the UK Used Car Dataset (Desai, 2021), a publicly available collection of $\sim 108,000$ listings scraped from a major UK automotive marketplace. The dataset provides a rich combination of numerical and categorical features, real-world missing values, and a heavily right-skewed continuous target (price in GBP), making it well-suited for exploring both modelling and preprocessing questions.

Research Questions. We address four specific research questions (RQs):

- RQ1 To what extent can machine learning models accurately predict used car prices from available features?
- RQ2 How do models spanning the bias-variance spectrum (linear models, instance-based, tree-based, and neural networks) compare in predictive performance and generalisation?
- RQ3 How do different data preprocessing strategies—target transformation, missing value imputation, and feature scaling—affect model performance?
- RQ4 Which features contribute most significantly to price prediction, and how can their importance be interpreted?

RQ1 establishes a baseline understanding of task difficulty. RQ2 provides a systematic model comparison linked to the bias-variance trade-off. RQ3 directly examines the downstream impact of preprocessing decisions, with particular relevance to distance-sensitive models such as KNN. RQ4 connects model be-

haviour to domain knowledge about the used car market.

The remainder of the paper is structured as follows: Section 2 reviews related work; Section 3 describes our dataset, preprocessing pipeline, models, and evaluation setup; Section 4 presents experimental results; Section 5 analyses findings; and Section 6 concludes.

2 Literature Review

Price prediction for used vehicles has attracted considerable attention in the machine learning literature. Pudaruth (2014) provided an early comparative study applying Naive Bayes, Decision Trees, and Neural Networks to a small Mauritian dataset, finding that tree-based methods outperform Naive Bayes due to the non-Gaussian nature of automotive features. Gegic et al. (2019) extended this to a larger European dataset using gradient-boosted trees and Support Vector Regression, demonstrating that ensemble methods consistently outperform linear baselines, with mileage and vehicle age emerging as the two most predictive features—a finding we replicate and contextualise in Section 4. Shinde and Kotecha (2018) proposed a KNN-based system for used car valuation, showing that KNN achieves competitive accuracy on datasets with strong local price clusters (e.g. vehicles of the same model and year), but degrades significantly without feature scaling—motivating our investigation of preprocessing effects in RQ3. Listiani (2009) applied SVR to German car listings, finding that RBF kernels outperform linear kernels, and that log-transforming the price target substantially improved RMSE.

Neural network approaches have gained traction with the availability of larger datasets. Goodfellow et al. (2016) provide the theoretical basis for multi-layer perceptrons as universal approximators, supporting their use for capturing non-linear feature interactions. Breiman (2001) introduced Random Forests, which have become a strong baseline for tabular regression due to their robustness to irrelevant features and insensitivity to feature scale.

Our work contributes to this literature by: (1) using a substantially larger and more diverse dataset than prior studies; (2) conducting a controlled comparison across five model

families; and (3) performing a systematic ablation of preprocessing strategies, which prior work treats as fixed.

3 Method

3.1 Dataset

We use the UK Used Car Dataset (Desai, 2021), a publicly available collection of used vehicle listings scraped from a major UK online automotive marketplace. The raw dataset comprises 108,540 listings distributed across 11 CSV files, one per manufacturer: Audi, BMW, Ford (two subsets), Mercedes (two subsets), Hyundai, Skoda, Toyota, Vauxhall, and Volkswagen. All files use the same nine-column structure: model, year, price in GBP, transmission, mileage, fuelType, tax as annual road tax in GBP, mpg as miles per gallon, and engineSize in litres. Two extra “unclean” files, Ford Focus and Mercedes C-Class, are also included as raw scraped data. These files contain issues such as malformed price strings and split mileage columns, so they are used to show real-world data quality problems and are not included in the main training dataset.

Properties. Table 1 summarises the key properties of the combined dataset after loading. The target variable price is continuous and exhibits severe right-skew (skewness = 2.29), with a median of £14,698 and values ranging from £450 to £159,999. Two features—tax (13.1% missing) and mpg (8.6% missing)—contain substantial proportions of missing values arising from incomplete scraping. The dataset contains 9 brands after merging the two Ford and two Mercedes subsets, and 195 distinct car models. Linear separability between price strata is low given the continuous target, but a moderate negative correlation with mileage ($r = -0.47$) and positive correlation with year ($r = 0.52$) provide a clear signal for regression.

3.2 Data Preprocessing

All preprocessing is implemented in Python using pandas and scikit-learn, and is fully reproducible via Notebook 01 in the accompanying code repository. The pipeline proceeds through five stages.

Property	Value
Total listings (raw)	108,540
Listings after cleaning	80,812
Number of brands	9
Number of models	195
Numerical features	6 (year, price, mileage, tax, mpg, engineSize)
Categorical features	3 (model, transmission, fuelType)
Target variable	price (continuous, GBP)
Target skewness (raw)	2.29
Target skewness (log)	-0.13
Missing values (tax)	13.1%
Missing values (mpg)	8.6%

Table 1: Summary statistics of the UK Used Car Dataset before and after cleaning. Listing counts are approximate post-cleaning due to duplicate and outlier removal.

Stage 1: Schema Standardisation. Each per-brand CSV is loaded and filtered to the nine standard columns, discarding any extraneous columns (e.g. the tax(GBP) column present in the Hyundai file due to a scraping inconsistency). All files are concatenated into a single DataFrame and a brand column is appended.

Stage 2: Validity Filtering. Four categories of physically implausible values are identified and treated as missing rather than deleted, preserving sample size:

- engineSize = 0: no internal combustion vehicle has zero displacement; likely a data-entry error.
- mileage = 0: used car listings with zero recorded mileage are treated as implausible data-entry defaults.
- price < £100: below any realistic transaction threshold.
- year $\notin$ [1990, 2023]: values of 1970 and 2060 represent obvious entry errors.

Stage 3: Duplicate Removal. Exact duplicate rows (identical across all nine columns) are removed.

Stage 4: Outlier Removal. We apply the IQR method with a conservative factor of $3\times$ to price, mileage, and tax. A row is removed if its value falls below $Q_1 - 3 \cdot \text{IQR}$ or above $Q_3 + 3 \cdot \text{IQR}$. This factor retains genuine high-value luxury cars and high-mileage fleet vehicles while discarding clear data errors.

The resulting price bounds are approximately £[1,229, 46,440].

Stage 5: Category Consolidation. Five fuel type labels and four transmission labels in the raw data include rare entries (e.g. Other, Electric for a small fraction of the dataset). We retain only {Petrol, Diesel, Hybrid, Electric} and {Manual, Automatic, Semi-Auto}, mapping all other values to NaN for subsequent imputation.

3.3 Missing Value Handling

Table 2 shows the missing value profile after validity filtering. We use group-median imputation for all numerical columns and within-brand mode imputation for categorical columns. Group-median imputation is preferred over global median because feature distributions are strongly stratified by brand and fuel type: for example, diesel engines have systematically higher displacement and different MPG profiles than petrol engines.

Feature	Imputation Strategy	Missing %
tax	Median by year and engine bin	~13%
mpg	Median by fuel type and engine bin	~9%
engineSize	Median by brand and fuel type	<1%
mileage	Median by year and brand	<1%
fuelType	Brand-level mode	<1%
transmission	Brand-level mode	<1%

Table 2: Missing value rates (post-cleaning) and imputation strategies. Engine bin refers to a coarse discretisation of engineSize into five intervals.

3.4 Feature Engineering

Beyond the nine raw features, we construct four derived features motivated by domain knowledge of vehicle depreciation:

- car_age = 2024 - year: vehicle depreciation is primarily a function of age rather than the registration year in isolation.
- mileage_per_year = mileage/(car_age + 1): normalises mileage by age, distinguishing a high-mileage new car from a low-mileage old one. A car driven 20,000 miles in one year signals different usage patterns than one driven 100,000 miles over ten years.

- `is_luxury` $\in \{0,1\}$: binary indicator for premium brands (Audi, BMW, Mercedes), which command a price premium independent of other features.
- `log_price` = $\log(1 + \text{price})$: log-transformation of the target to reduce right-skew (skewness from 2.29 to -0.13). All models are trained to predict `log_price`; results are exponentiated for interpretation.

3.5 Categorical Encoding

Notebook 01 creates several encoded helper columns for exploratory analysis. For model training in Notebook 02, those full-data encodings are excluded and categorical features are encoded inside the training pipeline to keep preprocessing consistent across train, development, and test splits. We apply the following encoding strategy:

- Median imputation is applied to numerical features inside each modelling pipeline.
- Most-frequent imputation is applied to categorical features inside each modelling pipeline.
- One-hot encoding (OHE) is applied to all categorical inputs: brand, model, transmission, and fuelType. This avoids imposing artificial ordinal relationships on manufacturer, model, fuel, or gearbox labels and allows unseen categories to be ignored safely at prediction time.

3.6 Feature Scaling

The final modelling feature set comprises the raw vehicle attributes plus engineered age, usage, and luxury-brand indicators (Table 3). We prepare two scaled versions for experiments related to RQ3:

- `StandardScaler`: zero mean, unit variance. Standard choice for gradient-based models and regularised linear models.
- `MinMaxScaler`: maps each feature to $[0, 1]$. Used to study the effect of bounded vs. standardised inputs on KNN and neural network performance.

Tree-based models (Decision Tree and Random Forest) are scale-invariant and are trained on unscaled numerical features.

Feature	Type	Source
<code>car_age</code>	Numerical	Engineered
<code>mileage</code>	Numerical	Raw
<code>mileage_per_year</code>	Numerical	Engineered
<code>tax</code>	Numerical	Raw
<code>mpg</code>	Numerical	Raw
<code>engineSize</code>	Numerical	Raw
<code>is_luxury</code>	Binary	Engineered
<code>brand</code>	Categorical	Raw, one-hot encoded in pipeline
<code>model</code>	Categorical	Raw, one-hot encoded in pipeline
<code>transmission</code>	Categorical	Raw, one-hot encoded in pipeline
<code>fuelType</code>	Categorical	Raw, one-hot encoded in pipeline
Target: <code>log_price</code>	Numerical	Engineered

Table 3: Final modelling feature set before categorical expansion. One-hot encoding is fitted inside each model pipeline.

3.7 Validation Setup

To enable fair comparison across all models and preprocessing variants, we adopt a unified validation protocol. The cleaned dataset is split into training (60%), development (20%), and held-out test (20%) sets using a fixed random seed (49). The test set is used only for final evaluation and is never seen during model selection or hyperparameter tuning.

Hyperparameter tuning is conducted on a fixed tuning subset from the training and development sets for runtime efficiency, while final development and test evaluation use the full corresponding splits. We additionally run a supplementary 3-fold cross-validation check on a 12,000-row training subset to assess whether the main model ranking is stable across folds. All models use the same splits and the same pipeline structure for fair comparison.

3.8 Models and Hyperparameters

We evaluate five models covering the bias-variance spectrum from high-bias/low-variance to low-bias/high-variance:

1. Ridge Regression: linear model with L2 regularisation; $\alpha \in \{0.1, 1, 10, 100\}$, best $\alpha = 0.1$.
2. K-Nearest Neighbours (KNN): instance-based distance model; $k \in \{5, 10, 20\}$

with uniform or distance weighting, best $k = 10$ with distance weighting.

3. Decision Tree: single non-linear tree model; maximum depth $\in \{8, 16, \text{None}\}$ and minimum leaf samples $\in \{1, 10, 50\}$, best no depth limit and leaf size 10.
4. Random Forest: bagged tree ensemble; trees $\in \{100, 200\}$, maximum depth $\in \{16, \text{None}\}$, and minimum leaf samples $\in \{1, 10\}$, best 200 trees with no depth limit and leaf size 1.
5. Multi-Layer Perceptron (MLP): feed-forward neural network with early stopping; hidden layers $\in \{(64), (128), (64, 32)\}$, $\alpha \in \{10^{-4}, 10^{-3}\}$, and learning rate = 0.001, best (64, 32) with $\alpha = 10^{-4}$.

3.9 Evaluation Metrics

All models are evaluated on the same three metrics, computed on original-scale predictions (after exponentiating `log_price` predictions):

- RMSE (Root Mean Squared Error): penalises large errors more heavily; primary metric.
- MAE (Mean Absolute Error): interpretable as average absolute price deviation in GBP.
- R^2 (Coefficient of Determination): proportion of price variance explained by the model.

For preprocessing comparisons (RQ3), we additionally report the cross-validation RMSE and its standard deviation to assess stability across folds.

4 Results

This section reports the model evaluation and optimisation results using the shared train–development–test split generated in Notebook 02. Model selection is based on the development set only; the held-out test set remains reserved for final evaluation. Since all models predict `log_price`, the reported price-scale metrics are computed after applying the inverse transformation to recover predicted prices in GBP.

4.1 Overall Model Comparison

Table 4 compares the five trained models on the development set. Random Forest achieves the best overall performance, with the lowest RMSE (GBP 1,866.08), the lowest MAE (GBP 1,217.09), and the highest price-scale R^2 (0.9541). This indicates that the ensemble model captures non-linear relationships between price and features such as transmission, year, engine size, mileage, and brand/model effects more effectively than the simpler baselines.

KNN is the second-best model on the development set, with RMSE of GBP 1,987.53 and R^2 of 0.9479. However, its training performance is almost perfect (training $R^2 = 0.9995$), suggesting substantial overfitting. MLP also performs competitively but does not improve over Random Forest or KNN. Decision Tree and Ridge Regression provide useful baselines, but their higher RMSE values show the limitation of single-tree and linear assumptions for this prediction task.

Model	MAE	RMSE	R^2
Random Forest	1217.09	1866.08	0.9541
KNN Regression	1286.03	1987.53	0.9479
MLP Neural Network	1412.49	2129.92	0.9402
Decision Tree	1433.38	2192.47	0.9366
Ridge Regression	1671.34	2444.19	0.9213

Table 4: Development set performance on the original price scale. Lower MAE/RMSE and higher R^2 indicate better performance.

Table 5 reports the final held-out test performance after model selection. The ranking remains consistent with the development results, confirming that the selected models generalise well beyond the tuning split. Random Forest again achieves the best performance, with a test RMSE of GBP 1,793.69 and a price-scale R^2 of 0.9570. KNN remains the second-best model, while MLP, Decision Tree, and Ridge Regression follow the same relative order as on the development set.

4.2 Hyperparameter Optimisation

The best hyperparameter settings selected during tuning are shown in Table 6. The selected Random Forest uses 200 trees and no fixed maximum depth, allowing the ensemble to model complex interactions while reducing

Model	MAE	RMSE	R^2
Random Forest	1205.22	1793.69	0.9570
KNN Regression	1274.86	1924.35	0.9505
MLP Neural Network	1410.22	2058.39	0.9433
Decision Tree	1421.53	2130.05	0.9393
Ridge Regression	1671.39	2410.53	0.9223

Table 5: Held-out test set performance on the original price scale.

variance through bagging. For KNN, distance weighting with $k = 10$ performs best, meaning closer neighbours contribute more strongly to the predicted price. Ridge Regression selects a small regularisation strength ($\alpha = 0.1$), while the MLP selects a two-layer architecture with 64 and 32 hidden units.

Model	Scaling	Best settings
Ridge Regression	Standard	$\alpha = 0.1$
KNN Regression	Standard	$k = 10$, distance weights
Decision Tree	None	leaf size = 10
Random Forest	None	200 trees, leaf size = 1
MLP Neural Net-work	Standard	layers (64, 32), $\alpha = 10^{-4}$

Table 6: Best hyperparameters selected for each model family.

To check whether the main model-comparison pattern is stable across different training folds, we also run a supplementary 3-fold cross-validation experiment on a 12,000-row sample from the training split. This experiment is used as a robustness check only; final model selection is still based on the development set and final reporting on the held-out test set.

Model	CV RMSE	CV R^2
Random Forest	2223.83 ± 77.55	0.9323
KNN Regression	2331.47 ± 39.54	0.9256
Ridge Regression	2480.74 ± 65.66	0.9158
Decision Tree	2870.28 ± 37.95	0.8872
MLP Neural Network	4266.30 ± 2368.32	0.6779

Table 7: Supplementary 3-fold cross-validation results on a 12,000-row training subset. RMSE is reported on the original price scale as mean $\pm$ standard deviation.

The cross-validation check supports the main conclusion for the strongest models: Random Forest and KNN remain the two best-performing methods. The lower-ranked models are less stable across folds, especially MLP,

whose large RMSE standard deviation indicates sensitivity to the sampled training data.

The optimisation results show a clear bias-variance pattern. Ridge Regression is stable but underfits because a linear function cannot fully represent the relationship between vehicle attributes and price. KNN has very low training error but a large train-development gap, showing high variance. Random Forest provides the best balance: its training error is lower than its development error, but the gap is much smaller than KNN, and its development and held-out test performance are the strongest overall.

4.3 Effect of Feature Scaling

Feature scaling has a strong effect on distance-based and linear models. Table 8 shows that Ridge Regression performs poorly without scaling (RMSE GBP 7,597.09), because features such as mileage and tax operate on much larger numeric scales than variables such as engine size. Standardisation and min-max scaling both substantially improve Ridge, reducing RMSE to approximately GBP 2,440.

KNN also benefits strongly from scaling. Without scaling, its development-tuning RMSE is GBP 6,889.90; after standardisation, this falls to GBP 2,097.92. Standard scaling outperforms min-max scaling for KNN in this experiment, likely because standardisation better preserves relative distances for skewed numerical variables.

Model	Scaling	RMSE	R^2
Ridge	None	7597.09	0.2393
Ridge	Standard	2444.76	0.9212
Ridge	Min-max	2442.11	0.9214
KNN	None	6889.90	0.3682
KNN	Standard	2097.92	0.9414
KNN	Min-max	2415.64	0.9223

Table 8: Scaling comparison for Ridge and KNN. KNN values are computed on the tuning subset for runtime efficiency.

4.4 Feature Influence

The Random Forest feature importance ranking provides insight into which attributes drive the strongest predictive signal. The most important feature is transmission_Manual, followed by year, engineSize, and car_age. This

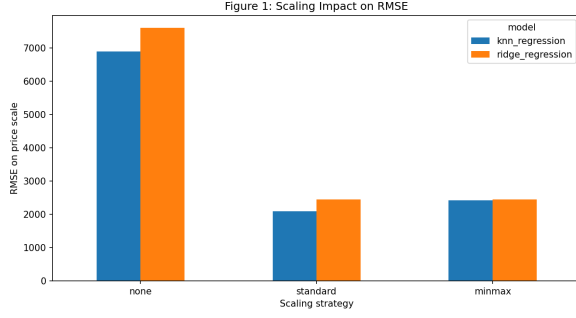


Figure 1: Effect of scaling strategy on RMSE for Ridge Regression and KNN.

ranking is consistent with used-car pricing intuition: newer cars, larger engines, and transmission type are strongly associated with market value. Mileage, MPG, and the luxury-brand indicator also contribute meaningful signal, although their individual importances are smaller than the top structural vehicle features.

Ridge coefficients provide a complementary linear interpretation. The largest absolute coefficients are mostly model-specific indicators, such as Caravelle, Up, KA, Aygo, and Supra. Positive coefficients indicate models associated with higher log-price after controlling for other variables, while negative coefficients indicate lower-price models. Because these coefficients are model-based associations, they should be interpreted as predictive signals rather than causal effects.

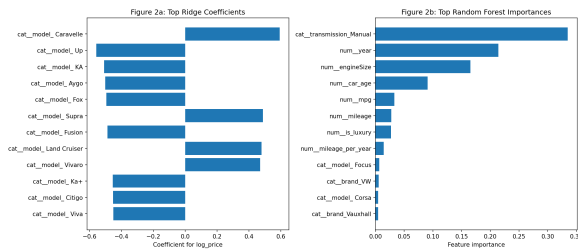


Figure 2: Top Ridge coefficients and Random Forest feature importances.

Overall, the evaluation supports the main research question, showing that vehicle attributes can predict used-car prices with high accuracy. The best-performing model was Random Forest, which explained about 95.7% of the variance in the held-out test set on the original price scale. It also achieved a lower prediction error than all baseline and neural network models.

5 Discussion

5.1 Model Behaviour Analysis

The test results show that Random Forest was the strongest model, with the lowest test RMSE of 1793.69 and the highest test R^2 of 0.9570. This may be because used car prices are affected by non-linear relationships and interactions between features such as brand, model, age, mileage, engine size, and other variables. Random Forest is able to capture these patterns, while its use of bagging reduces the variance of individual trees. As a result, it provides a good balance between model flexibility and generalisation.

KNN also performed well, with a test RMSE of 1924.35 and test R^2 of 0.9505, but it showed a clear overfitting risk. Its train RMSE was only 201.33, which is much lower than its test RMSE, suggesting that the model fitted local training samples too closely. Ridge Regression was the weakest model, with a test RMSE of 2410.53 and train RMSE of 2410.12, indicating underfitting caused by its limited linear structure. MLP learned non-linear patterns but did not outperform Random Forest, likely because tree-based models are better suited to this tabular dataset. The single Decision Tree was also weaker because it had higher variance than the ensemble model.

5.2 Error Analysis

Larger prediction errors are more likely for high-priced luxury cars, rare models, and vehicles with extremely low or high mileage or engine size. These cases are less common in the dataset, so the model has fewer similar examples to learn from. Their prices may also depend more strongly on individual condition, special options, or market demand.

Rare models are especially difficult after one-hot encoding because each category may only have a small number of samples. This makes it hard for the model to learn a stable pricing pattern for that specific model. In addition, the outlier removal process helped stabilise modelling in the mainstream price range, but it may also limit generalisation to very high-end vehicles.

Some remaining error is also caused by missing information. The dataset does not include variables such as location, trim level, accident

history, maintenance condition, seller type, or listing time. Listing price may also differ from the final sale price due to negotiation. Therefore, not all prediction error should be treated as a model weakness.

5.3 Preprocessing Impact

Scaling had a strong effect on Ridge Regression and KNN because both models are sensitive to feature scale. Without scaling, large-range variables such as mileage and tax can dominate the regularisation process or distance calculation, which leads to poor performance.

For Ridge Regression, the RMSE dropped from 7597.09 without scaling to 2444.76 with StandardScaler. MinMaxScaler gave a very similar result, with an RMSE of 2442.11. This shows that scaling is essential for regularised linear models in this dataset.

For KNN, the effect was even clearer. The RMSE decreased from 6889.90 without scaling to 2097.92 with StandardScaler, while MinMaxScaler achieved 2415.64. Since KNN defines similarity through distances, scaling is necessary to make the distance between cars meaningful. The KNN scaling comparison was conducted on a tuning subset for runtime efficiency, so these values are mainly used to compare preprocessing choices. Overall, preprocessing directly affected whether these models could learn useful patterns.

5.4 Feature Importance Interpretation

The Random Forest feature importance results show that the main predictors were `transmission_Manual`, `year`, `engineSize`, `car_age`, `mpg`, `mileage`, and `is_luxury`, with importance scores of 0.3347, 0.2143, 0.1653, 0.0910, 0.0330, 0.0276, and 0.0273 respectively. These features match common used-car pricing logic because they reflect depreciation, usage level, vehicle class, and brand premium.

`year` and `car_age` represent depreciation, while `mileage` reflects how much the vehicle has been used. Engine size and transmission are linked to vehicle specification and market positioning, and the `is_luxury` indicator captures the price premium associated with luxury brands. Ridge coefficients also showed that specific model identity has a strong predictive association with price, with positive

coefficients for models such as Caravelle and negative coefficients for smaller economy models such as Up, KA, and Aygo. These results should be interpreted as predictive associations rather than causal effects.

5.5 Limitations and Future Improvements

This study has several limitations. First, the target variable is listing price rather than final transaction price, so it may include seller expectations, negotiation space, and pricing strategy. Second, the dataset lacks important variables such as location, vehicle condition, accident history, trim level, seller type, and listing date. These missing factors may explain part of the remaining prediction error.

Third, the data comes from the UK used-car market, so the findings should not be directly generalised to other countries or regions. Market conditions, brand preferences, tax rules, and resale patterns may vary across different locations. The modelling process also had some limitations. Due to runtime and course-scope constraints, cross-validation was only used as a robustness check on a 12,000-row subset.

Future work could add more detailed vehicle and market information, followed by a comparison between Random Forest and other strong tabular models such as Gradient Boosting, XGBoost, or LightGBM. SHAP values or permutation importance could also be applied to assess whether the main predictors remain stable across different model settings.

6 Conclusion

RQ1. Used car prices can be predicted effectively using the available structured vehicle features. The best-performing model was Random Forest, with a test RMSE of 1793.69, MAE of 1205.22, and R^2 of 0.9570. This means that it explained around 95.7% of the price variance in the held-out test set.

RQ2. The model comparison shows that non-linear models performed better than the linear baseline, although their generalisation ability varied across model types. Random Forest achieved the best test performance, with an RMSE of 1793.69 and R^2 of 0.9570, reducing test RMSE by 25.6% compared with Ridge Regression. KNN was the second-best

model on the test set, but its much lower train RMSE of 201.33 compared with its test RMSE of 1924.35 suggests high variance and overfitting. MLP also performed reasonably well, but it did not outperform Random Forest. The single Decision Tree was weaker than the ensemble model, indicating that bagging helped improve both stability and accuracy.

RQ3. Preprocessing had a clear impact, especially for models that are sensitive to feature scale. Standard scaling reduced Ridge RMSE from 7597.09 to 2444.76, and KNN RMSE from 6889.90 to 2097.92. This shows that large differences in the ranges of mileage, tax, and engine size can affect linear regularisation and distance-based calculations. Missing-value imputation helped retain usable training samples, while one-hot encoding allowed categorical variables such as brand, model, fuel type, and transmission to provide useful predictive information. The log-price transformation also reduced target skewness from 2.29 to -0.13 , making the regression task more stable.

RQ4. The most important predictor was `transmission_Manual`, with a Random Forest importance score of 0.3347. This was followed by `year` at 0.2143, `engineSize` at 0.1653, and `car_age` at 0.0910. These features reflect differences in vehicle specification, depreciation, and market segment. In general, newer cars and cars with larger engines tend to have higher prices, while transmission type and luxury-brand status capture differences in vehicle class and buyer demand.

Overall, Random Forest is the best choice for prediction accuracy in this experiment, while future improvement should focus on richer vehicle-level data, gradient boosting models, and stronger interpretability checks.

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